

PROJECT PROPOSAL

CS3631 – Deep Neural Networks · Research Project (2026)

Does Efficiency Cost Robustness? A Lung-Supervised, Explainable, From-Scratch CNN for Chest X-ray Screening

Investigating the efficiency-robustness-faithfulness trade-off in compact chest X-ray models, and using lung-region supervision to break it — for deployment in low-resource clinics.

Research objective	Optimize a DNN (§3-3): efficient + robust + explainable
Contribution type	Efficiency + learning strategy + explainability framework (§8)
Baseline (§4)	Standard CNN, trained from scratch
Proposed model	LiteCXR-Net (from scratch, lung-supervised, no pretrained weights)
Primary dataset	COVID-19 Radiography Database (Kaggle, §2.2)
Compute	Free Google Colab / Kaggle GPU (§6)
Team	5 members — each owns a hands-on ML task
Date	July 2026

Contents

1. Abstract

Automated chest X-ray (CXR) screening is most needed in low-resource clinics, which run on modest hardware — yet the most accurate models are large, slow, and, worse, untrustworthy: they frequently exploit shortcut learning, keying on source-hospital or scanner artefacts rather than lung pathology, so their accuracy collapses on data from a new hospital. Making models smaller for deployment raises an unresolved question: does compressing a CXR model make shortcut learning better or worse?

This project answers that question and acts on it. We first quantify the **efficiency-robustness-faithfulness trade-off** for compact CNNs trained entirely from scratch (no pretrained weights). We then propose **LiteCXR-Net**, a lightweight architecture trained with explicit **lung-region supervision** designed to break that trade-off — remaining accurate and small while being robust to source shortcuts and anatomically faithful. A controlled comparison against an identical model trained without lung supervision isolates the causal effect of the recipe, and a lung-localisation faithfulness score built on Grad-CAM turns interpretability into a measurable outcome. All experiments run on free GPUs, and every design decision is justified through hyper-parameter tuning, ablations and computational analysis.

2. Problem Motivation

Chest diseases such as pneumonia, COVID-19, tuberculosis and lung opacity are among the leading causes of death worldwide. Radiologists diagnose them from chest X-rays, but interpretation is slow and expertise is scarce, especially in low-resource regions. Automated triage is valuable — but only if it is both deployable and trustworthy.

2.1 Two coupled constraints

Resource cost. The most accurate published CXR models have tens of millions of parameters and high inference cost, impractical for a single CPU, a low-end GPU, or a mobile device — exactly the hardware of the clinics that most need screening.

Trust / shortcut learning. Publicly compiled CXR datasets often draw each class from a different source, so a model can score highly by recognising the source rather than the disease; such models fail on new hospitals and cannot be trusted clinically.

These constraints are usually treated in isolation. In practice they interact: the act of shrinking a model for deployment may change how much it relies on shortcuts — and that interaction is what this project investigates and exploits.

3. Research Gap & Central Hypothesis

A focused reading of recent work (efficient medical imaging, shortcut learning, and interpretability) reveals three gaps that together define a genuine, publishable question.

3.1 What is missing

- **Efficient CXR models almost always start from large pretrained backbones.** This imports ImageNet biases and hides the true cost and behaviour of a small model that must learn from the medical data alone. Genuinely from-scratch, compact CXR models are under-studied.
- **The efficiency-robustness interaction is unresolved.** Shortcut learning is documented for large models, but its relationship to model capacity is not. Two opposing hypotheses coexist and neither has been tested

on chest X-rays: (a) lower-capacity models latch onto the easiest, most spurious cues more strongly, so compression hurts robustness; (b) compression acts as a regulariser, so smaller models are more robust. Which holds for CXR is unknown.

- **Faithfulness is rarely quantified or linked to the trade-off.** Explainability in CXR is usually a qualitative Grad-CAM figure. Almost no work measures how much of a model's attention lies inside the lungs, nor connects that faithfulness to efficiency and robustness.

3.2 The gap, stated precisely

Research gap

It is unknown whether a compact, from-scratch CNN for chest X-rays must trade robustness (and anatomical faithfulness) for efficiency — and whether an explicit lung-region training signal is the lever that removes that trade-off, yielding a model that is simultaneously small, robust to source shortcuts, and faithful.

3.3 Central hypotheses

ID	Hypothesis
H1 (the trade-off)	Naively compressed from-scratch CNNs suffer a larger internal-to-external (out-of-distribution) accuracy drop than a larger baseline — i.e., efficiency, obtained naively, worsens shortcut reliance.
H2 (the lever)	Training a compact model with explicit lung-region supervision (LiteCXR-Net) reduces the out-of-distribution drop and raises the lung-faithfulness score at similar or lower cost — breaking the efficiency-vs-robustness trade-off.

Why this is likely to be accepted

It poses a clear, previously-unanswered question (does efficiency cost robustness in CXR?), not another benchmark.

It is causal and controlled: a recipe-vs-identical-control comparison isolates the effect of lung supervision, which reviewers value far more than a single leaderboard number.

It ties three properties reviewers care about — efficiency, robustness, interpretability — with a measurable faithfulness metric, and it is fully reproducible on free hardware.

It has a concrete deployment motivation (low-resource clinics), giving the 'why it matters' that pure accuracy papers lack.

4. Research Objectives

1. Implement a standard from-scratch CNN baseline and characterise its accuracy, efficiency, robustness and faithfulness (mandatory baseline, §4).
2. Empirically map the efficiency–robustness–faithfulness trade-off by measuring how out-of-distribution robustness and lung-faithfulness change as the model is compressed (width multiplier sweep).
3. Design LiteCXR-Net — a compact, from-scratch CNN trained with lung-region supervision — intended to break that trade-off.
4. Isolate the causal effect of lung supervision through a controlled comparison against an identical model trained without it.
5. Validate every design decision with hyper-parameter tuning, ablation studies, computational analysis and a quantitative faithfulness score, packaged as a reproducible, conference-ready study.

5. Research Questions

ID	Research question	Tests
RQ1	Can a from-scratch compact CNN match a standard baseline's accuracy at a fraction of the computational cost?	Efficiency
RQ2	Do compact CNNs trade out-of-distribution robustness for efficiency (is there a trade-off)?	H1 (core)
RQ3	Does lung-region supervision break the trade-off — small AND robust AND faithful?	H2 (core)
RQ4	Which architectural and training components drive the accuracy–efficiency–robustness balance?	Ablation

6. Proposed Contribution & Novelty

C1 — Scientific finding (empirical novelty). The first systematic characterisation of the efficiency–robustness–faithfulness trade-off for from-scratch CNNs on chest X-rays, answering whether compression helps or hurts shortcut learning.

C2 — LiteCXR-Net (architecture + learning-strategy novelty, §8). A compact architecture combined with a lung-region-supervised training recipe that breaks the efficiency-vs-robustness trade-off, delivering baseline-level accuracy at a fraction of the cost with a smaller out-of-distribution drop.

C3 — Faithfulness framework (explainability, §8). A lung-localisation faithfulness score, evaluated with a controlled recipe-vs-control ablation that causally links lung supervision to both robustness and interpretability.

C4 — Reproducible, deployment-ready artefact. An open, from-scratch model and protocol that runs on free hardware, with practical guidance for low-resource clinical deployment.

7. Dataset

Primary — COVID-19 Radiography Database (Kaggle, §2.2). Four classes (COVID-19, Normal, Lung Opacity, Viral Pneumonia), ~21,000 frontal chest X-rays. Its documented source bias is an asset here: it is the very shortcut phenomenon our recipe targets. Class imbalance (Normal ~10.2k · Lung Opacity ~6.0k · COVID ~3.6k · Viral Pneumonia ~1.3k) is handled with class weighting.

External / robustness set. Trained models are evaluated — without retraining — on an external source (RSNA Pneumonia Detection Challenge, mapped to a shared label space), or a source-stratified hold-out if single-dataset use is preferred. This external evaluation is the measurement instrument for the robustness (H1/H2) study.

8. Baseline Model (Mandatory, §4)

The first deliverable is a standard from-scratch CNN (e.g. a ResNet-18-style stack of Conv–BatchNorm–ReLU residual blocks with a global-average-pooling head and 4-way softmax), with no pretrained weights. Inputs are 224×224 (128×128 fast-mode for tuning), grayscale-to-3-channel, normalised; training uses class-weighted cross-entropy, AdamW, a cosine/step schedule, early stopping and a fixed seed, in PyTorch. For the proposal deliverable we report its accuracy, macro-F1, ROC-AUC, parameter count / FLOPs / latency, and a hyper-parameter tuning sweep. It is the reference for every later comparison.

9. Proposed Framework: LiteCXR-Net

9.1 Efficient architecture

- **Depthwise-separable convolutions** cut parameters and FLOPs by roughly an order of magnitude per layer while preserving representational power.
- A **width multiplier α (0.5 / 0.75 / 1.0)** is an explicit efficiency dial — and the instrument we sweep to map the efficiency–robustness trade-off (H1).
- A **lightweight channel-attention block** recovers accuracy lost to the compact design, at negligible cost.
- A **compact head** (global average pooling + one linear layer) keeps the classifier small and reduces overfitting. Target budget: roughly 0.5–2 M parameters.

9.2 Lung-region supervision (the lever, trained via a U-Net)

A lightweight U-Net lung-segmentation model is trained to produce lung masks. These masks drive the shortcut-mitigation recipe in two ways:

- **Input masking / lung-focused cropping** removes borders, text markers and corner artefacts that carry source information, forcing the model to use lung content.
- A **lung-focus training signal** (strong augmentation plus an optional penalty on activation outside the lung mask) discourages reliance on non-lung cues. This is the recipe hypothesised to break the trade-off (H2).

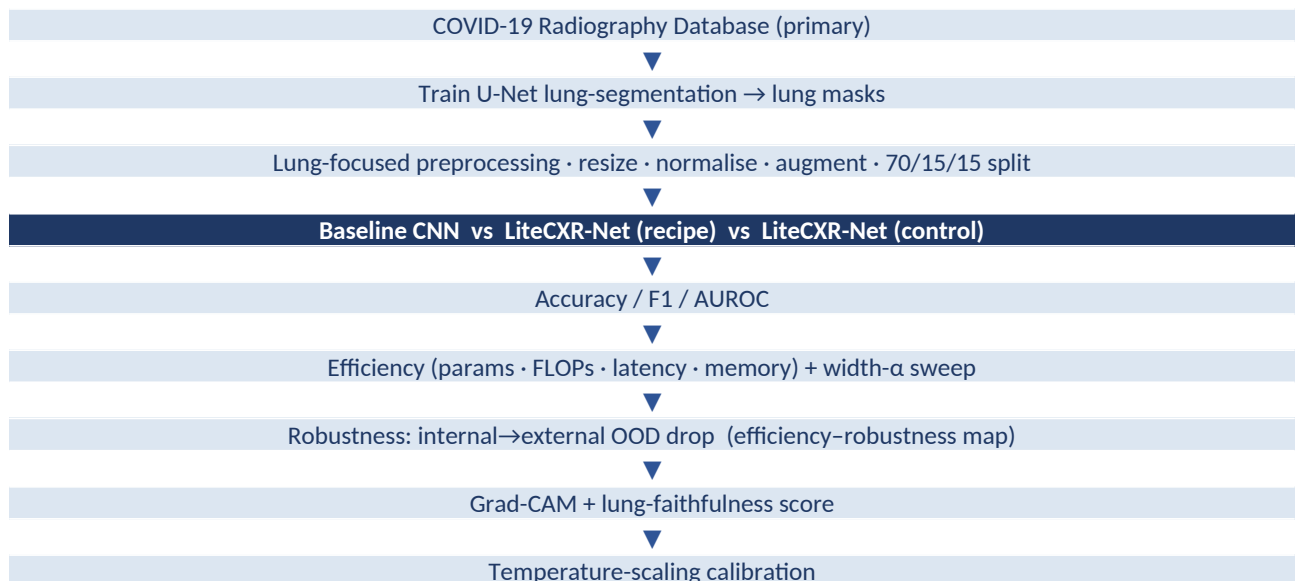
9.3 Controlled evaluation of the recipe

To attribute any robustness gain to lung supervision rather than to the architecture, we train two otherwise-identical LiteCXR-Net models: one with the lung-supervision recipe and one control without it. The difference in their out-of-distribution drop and faithfulness score is the causal effect of the recipe.

9.4 Explainability & faithfulness

Grad-CAM heatmaps are computed from the last convolutional layer; the **lung-localisation faithfulness score** is the fraction of Grad-CAM activation mass inside the U-Net lung mask. It measures, numerically, whether predictions are anatomically grounded — the same masks thus serve both training and evaluation.

9.5 End-to-end pipeline



▼

Hyper-parameter tuning + ablations → paper

10. Methodology & Implementation

10.1 Preprocessing

Remove corrupted/duplicate images; generate U-Net lung masks; lung-focused crop/mask; resize (224 or 128 fast-mode); grayscale-to-3-channel; normalise; stratified 70/15/15 split with fixed seed and saved indices.

10.2 Training protocol (shared)

Baseline, LiteCXR-Net (recipe) and LiteCXR-Net (control) use the same optimiser, schedule family, class weighting, early stopping and seed, so differences are attributable to the architecture and the lung-supervision recipe — not to tuning luck. Mixed precision keeps memory within free-GPU limits.

10.3 Reproducibility

GitHub version control, fixed seeds, file-based configs, full metric logging, and runnable code shipped with the paper.

11. Experimental Plan

11.1 Experiments

#	Experiment	Purpose
E1	Baseline vs LiteCXR-Net — accuracy, macro-F1, ROC-AUC	Comparative (RQ1)
E2	Hyper-parameter tuning (LR, batch, width α , weight decay, aug strength)	Mandatory HP tuning
E3	Ablation: depthwise-separable vs standard conv; $\pm$ attention; $\pm$ lung recipe; α sweep	Mandatory ablation (RQ4)
E4	Computational analysis: parameters, FLOPs, latency (CPU/GPU), memory, training time	Efficiency (RQ1)
E5	Efficiency-robustness map: OOD drop across width α (compression levels)	H1 — the core question (RQ2)
E6	Recipe vs identical control: causal effect of lung supervision on OOD drop & faithfulness	H2 (RQ3)
E7	Explainability: Grad-CAM + lung-faithfulness score, baseline vs proposed	Interpretability (RQ3)
E8	Calibration: temperature scaling + reliability (supporting)	Trust (supporting)

11.2 Evaluation metrics

Category	Metrics
Classification	Accuracy, Precision, Recall, macro-F1, per-class F1, ROC-AUC, confusion matrix
Efficiency	Parameters, FLOPs, model size (MB), latency (CPU & GPU), peak memory, training time
Robustness	Internal-to-external accuracy & AUROC drop (generalisation gap), across width α
Faithfulness	Lung-localisation faithfulness score; qualitative heatmaps
Calibration (support)	Expected Calibration Error (ECE), reliability diagram

12. Computational Feasibility (§6)

All experiments target free Google Colab / Kaggle GPUs. The U-Net and the classification models are small; at 224×224 with mixed precision a full run takes on the order of tens of minutes, and the 128×128 fast-mode makes the width- α sweep and tuning cheap. No multi-GPU, HPC or paid cloud is required; frequent checkpointing guards against session limits.

13. Rules Compliance (CS3631)

Rule	Compliance
§2 Dataset	COVID-19 Radiography Database (Kaggle) — permitted under §2.2.
§3 Objective	Objective 3 — optimise a DNN (size/latency/memory) with improved robustness.
§4 Baseline	Standard CNN, trained from scratch, fully evaluated.
§5 Model constraints	CNN + U-Net, all trained from scratch — no pretrained/foundation models, no LLM/agent.
§6 Compute	Runs on free Colab/Kaggle GPU.
§8 Contribution	Efficiency + learning-strategy (lung supervision) + explainability framework.
§9 Experiments	Baseline comparison, HP tuning, ablation, computational analysis — all included.
§11 Individual work	Every member owns a hands-on ML task — a model they train/fit (see §15).

14. Timeline (One Month)

The core research completes within one month. Note the official milestones: 2-page ACM proposal (Aug 2), short paper (Aug 23), full paper (Week 12); the Week-1 baseline results feed the ACM proposal.

Week	Focus	Output
1	Data pipeline + U-Net lung masks; from-scratch baseline CNN + HP tuning	Baseline + efficiency numbers (feeds ACM proposal)
2	Train LiteCXR-Net (recipe) and control; baseline comparison; width- α models	Proposed & control models trained
3	Efficiency-robustness map + ablations + Grad-CAM faithfulness + calibration	Full results and figures
4	Consolidate, write paper, prepare submission	Conference-ready paper + code

15. Team Roles (§11 — each owns a hands-on ML task)

Every member trains or fits at least one model, satisfying the requirement that each student contributes to an identifiable DNN stage, not merely writing.

Member	Hands-on ML task (trains/fits a model)	Also owns	Paper section
M1	Trains & tunes the from-scratch baseline CNN	Repo, tracking, seeds, references	Intro, Related Work, Abstract
M2	Trains the lung-segmentation U-Net + the lung-supervised (recipe) model	Data pipeline + preprocessing	Dataset, Experimental Setup
M3	Designs & trains LiteCXR-Net + hyper-parameter tuning	Proposed architecture	Proposed Framework
M4	Trains & evaluates all ablation model variants (α sweep, $\pm$ attention, $\pm$ conv)	Efficiency benchmark + Pareto	Experiments (efficiency, ablation)
M5	Trains the no-recipe control model + fits temperature-scaling calibration	Robustness + Grad-CAM/faithfulness + assembly	Results, Discussion, Conclusion

16. Risks & Mitigations

Risk	Mitigation
From-scratch training under-fits	Strong augmentation, class weighting, moderate baseline capacity; optional self-supervised pre-training on CXR (no ImageNet).
U-Net masks imperfect	Use masks as soft guidance; fall back to intensity-based lung cropping; masks need

Risk	Mitigation
	not be pixel-perfect for the faithfulness score.
Recipe shows no robustness gain	Still a valid finding (reported via the controlled ablation); the efficiency contribution and the trade-off map stand alone.
External dataset formatting differs	Map to a shared label space; fall back to a source-stratified hold-out.
Free-GPU limits	128×128 fast-mode, mixed precision, frequent checkpoints.

17. Expected Contributions

6. An empirical map of the efficiency–robustness–faithfulness trade-off for from-scratch CXR CNNs (answers: does compression help or hurt shortcut learning?).
7. LiteCXR-Net: a compact, lung-supervised model that breaks the trade-off — accurate, small, robust, and faithful.
8. A causal, controlled demonstration (recipe vs identical control) linking lung supervision to robustness and interpretability.
9. A quantitative lung-localisation faithfulness score and a reproducible, free-hardware protocol with open code.
10. Deployment guidance for accurate, trustworthy CXR screening on modest hardware.

18. Indicative References

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